HANDS-ON ACTIVITY FOCUS GUIDE

Mix and Normalize Split Arcade Drive

One stick drives, one stick turns

25 MIN	8	MATH -> ROBOT	1
ACTIVITY	STUDENT ROLES	FORMAT	SAFETY LEAD

ACTIVITY OUTCOME Students combine drive and turn inputs, recognize an out-of-range mix, normalize both outputs with the same denominator, and connect the result to robot motion.

Materials

- BasicArcadeDriveTeleOp.java or the five key split-arcade lines displayed.
- Role cards: DRIVE, TURN, ADD, SUBTRACT, DENOMINATOR, LEFT POWER, RIGHT POWER, and MOTION.
- Value cards including 0.0, +0.2, +0.5, +0.6, and +1.0.
- Printed mix-and-normalize worksheet from this guide.
- Optional secured robot with Drive, Turn, Left Power, and Right Power telemetry.

Before students enter

- Confirm the convention: positive drive requests forward; positive turn requests a right turn for this example.
- Write the left and right formulas where every student can see them.
- Prepare at least one case that does not need normalization and one that does.
- Keep the robot disabled until students complete the math cases successfully.

SAFETY BOUNDARY Students never cross into the robot test area while the OpMode is enabled. STOP and visually confirm DISABLED before anyone approaches.

Student setup and roles

Assign one primary role to each student. Rotate roles on a second trial if time permits. The safety lead may stop the activity at any time.

STUDENT	ROLE	RESPONSIBILITY
1	DRIVE	Holds -gamepad1.left_stick_y.
2	TURN	Holds gamepad1.right_stick_x.
3	ADD	Calculates drive + turn.
4	SUBTRACT	Calculates drive - turn.
5	DENOMINATOR	Calculates max(abs(drive)+abs(turn), 1.0).
6	LEFT POWER	Divides the left raw mix by the denominator.
7	RIGHT POWER	Divides the right raw mix by the denominator.
8	MOTION	Predicts the resulting robot movement.

Ground rules

- Predict before testing; record evidence before proposing a correction.
- Only the Driver Station operator touches INIT, PLAY, or STOP.
- Any student may call STOP for unexpected motion, instability, heat, noise, or an unclear boundary.
- Change only one variable between tests so the evidence remains interpretable.

POWERING ON gate

SAY AND DO ALL SIX STEPS 1. Secure the robot. 2. Confirm DISABLED. 3. Look around. 4. Make sure everyone is clear. 5. Call POWERING ON! and wait. 6. Visually confirm clear, then enable/PLAY.

After every test: press STOP, visually confirm DISABLED, and only then permit anyone to approach the robot.

25-minute teacher walkthrough

1. Compare control styles | 3 minutes

TEACHER	*"Ask students to show with their hands how tank drive and split arcade assign joystick jobs."*

STUDENT	*"Students state: tank uses one Y stick per side; split arcade uses left Y for drive and right X for turn."*

ASK	*"Why is this called split arcade?"*

LISTEN FOR The forward/back and turning controls are split across two sticks.

TEACHING NOTE Do not call it single-stick arcade; that usually places both axes on one stick.

2. Mix a straight command | 4 minutes

TEACHER	*"Set drive = +0.5 and turn = 0.0. Walk through raw left, raw right, denominator, and final outputs."*

STUDENT	*"ADD and SUBTRACT both produce +0.5; DENOMINATOR produces 1.0; both power roles show +0.5; MOTION predicts straight forward."*

ASK	*"Why does division by 1.0 leave the outputs unchanged?"*

LISTEN FOR Values already in range should not be amplified or reduced.

3. Mix a pure turn | 4 minutes

TEACHER	*"Set drive = 0.0 and turn = +0.5. Require both formulas to be evaluated."*

STUDENT	*"Left raw mix is +0.5; right raw mix is -0.5; denominator is 1.0; MOTION predicts a right in-place turn under this convention."*

ASK	*"Why do the formulas use opposite turn signs?"*

LISTEN FOR One side must increase relative to the other to create steering.

TEACHING NOTE Connect the sign pattern to the earlier tank-drive motion lesson.

4. Normalize an out-of-range mix | 7 minutes

TEACHER	*"Set drive = +1.0 and turn = +1.0. Do not skip the raw-mix step."*

STUDENT	*"ADD shows +2.0; SUBTRACT shows 0.0; DENOMINATOR shows 2.0; final powers become +1.0 and 0.0."*

ASK	*"Why divide both sides by the same number?"*

LISTEN FOR It brings both into range while preserving the ratio that creates the intended steering.

TEACHING NOTE Normalization is not clamping each side independently; independent clipping can distort steering.

5. Student case and safe transfer | 7 minutes

TEACHER	*"Students solve drive = +0.6, turn = +0.2. After checking +0.8/+0.4 with denominator 1.0, optionally transfer to the secured robot."*

STUDENT	*"Students predict a gentle right curve. For a live test, the team completes POWERING ON, tests one small command, presses STOP, and confirms DISABLED."*

ASK	*"What should telemetry show for the solved case?"*

LISTEN FOR Drive +0.6, Turn +0.2, Left Power +0.8, Right Power +0.4.

TEACHING NOTE Use smaller physical stick inputs than the full-value paper examples.

Split-arcade mixing and normalization

Student/team: ________________________________ Date: ______________ Robot: __________________

DRIVE / TURN	RAW LEFT / RIGHT	DENOMINATOR	FINAL L / R + MOTION
+0.5 / 0.0			
0.0 / +0.5			
+0.6 / +0.2			
+1.0 / +1.0			
-0.5 / 0.0			
Student case: ____ / ____			

STUDENT EVIDENCE RULE Write what was commanded and what was observed. Do not write a correction until STOP has been pressed and DISABLED has been confirmed.

Safety sign-off

Before live testing, the safety lead confirms: robot secure [ ] DISABLED [ ] boundary clear [ ] POWERING ON called and wait completed [ ]

After testing: STOP pressed [ ] DISABLED visually confirmed [ ] Safety lead initials: __________

COMPLETION CRITERIA The student uses lesson vocabulary, predicts before testing, records evidence, explains one result from the code or math, and follows STOP/DISABLED rules before approaching the robot.

Debrief

ASK	LISTEN FOR
What are the two driver inputs?	Left-stick Y supplies drive after negation; right-stick X supplies turn.
Why can raw mixes reach magnitude 2.0?	Drive and turn can each approach magnitude 1.0 and are added on one side.
What does the denominator guarantee?	It is at least 1.0 and scales out-of-range mixes back into the motor-power range.
What relationship must normalization preserve?	The ratio between left and right outputs, which preserves steering intent.

Troubleshooting without guessing

OBSERVATION	NEXT CHECK AFTER STOP/DISABLED
Robot turns opposite the lesson prediction	After STOP/DISABLED, verify turn sign and motor direction configuration.
One output exceeds 1.0 on paper	Calculate abs(drive)+abs(turn), use at least 1.0, then divide both raw outputs.
Student divides only the larger side	Require one shared denominator for both left and right outputs.
Straight command curves	Check direction, mechanical drag, wheel condition, and side-to-side output evidence.

Technical notes for the teacher

- This is split/two-stick arcade drive; single-stick arcade uses both axes of one stick.
- leftPower = (drive + turn) / denominator; rightPower = (drive - turn) / denominator.
- max(abs(drive) + abs(turn), 1.0) avoids amplifying ordinary in-range values.
- Dividing both sides by the same denominator preserves the steering ratio.
- The required reversed motor side depends on the physical drivetrain, not the drive style's name.

EXIT TICKET For drive = +0.6 and turn = +0.2, calculate leftPower and rightPower and state whether normalization is needed.